INCH-POUND

MIL-M-52590B(ME)
17 October 1991
SUPERSEDING
MIL-M-52590A(ME)
11 January 1983

MILITARY SPECIFICATION

MUFFLERS, EXHAUST, INTERNAL COMBUSTION

ENGINE, GENERAL PURPOSE

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers exhaust mufflers for general purpose application on internal combustion engines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 2990

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-14105	- Paint, Heat-Resisting (For Steel Surfaces).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Unless otherwise indicated, copies of federal and military specifications and handbooks are available from the Standardization Documents Order Desk, Building. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 176	- Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.
A 463	- Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1 and Type 2.
D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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AMERICAN WELDING SOCIETY (AWS)

D1.3	- Structural Welding Code - Sheet Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135-3030.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J366	- Exterior Sound Level for Heavy Trucks and Buses.
J1174	- Operator Ear Sound Level Measurement Procedure for Small Engine Powered Equipment.
J1207	- Measurement Procedure for Determination of Silencer Effectiveness in Reducing Engine Intake or Exhaust Sound Level.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The muffler shall be a special duct or pipe that impedes the transmission of sound while permitting the flow of exhaust gases from internal combustion engines.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. The material shall be stainless and heat-resisting chromium steel in accordance with ASTM A 176, UNS S40900 or aluminum-coated steel in accordance with ASTM A 463, type 1, T1-40 coating designation. Materials not specified shall be selected by the contractor, and shall be subjected to all provisions of this specification. When tested in accordance with 4.5.2.4, the loss in weight of material shall not be more than 8 grams per square foot or surface area.

3.3.1 Material deterioration prevention and control. The muffler shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the muffler may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make this information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the muffler may be newly fabricated from recovered materials to the maximum extent practicable, provided the muffler produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the muffler.

3.4 Noise reduction. The mufflers shall reduce engine exhaust noise (insertion loss) a minimum of 25 dB(A) below the unsilenced noise of the engine when tested in accordance with 4.5.2.1.

3.5 Back pressure. The muffler shall not create a back pressure of more than 8 inches of water for engines up to and including 10 horsepower, and not more than 20 inches of water for engines over 10 horsepower.

3.6 Resistance to air leakage. The muffler shall not leak more than 3-1/2 cubic feet per minute (cfm) of air when subjected to internal air pressure of 4.5 pounds per square inch (psi) pressure.

3.7 Welding. When specified (see 6.2), the end caps and shell shall be welded in accordance with AWS D1.3. Welding procedure specifications and welders and welding operators shall be qualified in accordance with AWS D1.3.

3.8 Interchangeability. All mufflers having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Identification marking. The mufflers shall be identified in accordance with MIL-STD-130, method optional. In addition, the muffler inlet shall be clearly indicated.

3.10 Treatment and painting. When specified (see 6.2), the muffler shall be cleaned, treated, and painted in accordance with MIL-T-704 with paint conforming to MIL-P-14105, color optional. Stainless steel surfaces shall be properly prepared and roughened prior to painting.

3.11 Workmanship. The mufflers shall be free from sharp bends and edges, defective joints, and other defects which would affect the serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. A first article shall be performed on one complete muffler.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The muffler shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable operating and storage environment (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the muffler (see 3.3.2).
106. Welding procedures not qualified as specified (see 3.7).
107. Welders not qualified as specified (see 3.7).
108. Mufflers not functionally and dimensionally interchangeable.
109. Identification marking not as specified (see 3.9).
110. Muffler inlet not clearly indicated (see 3.9).
111. Treatment and painting not as specified (see 3.10).
112. Workmanship not as specified (see 3.11).

4.5.2 Tests.

4.5.2.1 Noise level test. The insertion loss of the mufflers shall be determined by measuring the silenced and unsilenced engine. The engine shall be acoustically isolated from the exhaust opening. Noise levels shall be measured in accordance with SAE J1207, except that measurements shall be made at a horizontal distance of five feet from the opening of the muffler (silenced engine) or pipe (unsilenced engine). In addition, noise measurement data recording shall be in accordance with MIL-STD-1474, paragraph 5.5. Nonconformance with 3.4 shall constitute failure of this test.

4.5.2.2 Back pressure. The exhaust back pressure shall be measured by a manometer. The point of measurement shall be no more than 2 inches beyond the exhaust manifold outlet flange at rated load. The measurement shall be taken on a neutral flow plane at least one pipe diameter away from any changes in flow direction. Not less than 2 tests shall be made and the results averaged. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.3 Leakage. The inlet and outlet of the muffler shall be sealed. A means shall be provided to permit pumping 4.5 psi of air into the muffler. Nonconformance to 3.6 shall constitute failure of this test.

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4.5.2.4 Corrosion resistance. The following corrosion resistance test shall be performed on samples of material which are identical with the material used in the mufflers.

1. Two test panels, each 2- by 3-inches shall be used for this test. A 3/16-inch hole shall be punched near one 2-inch side of each panel. The panels shall be suspended by a glass rod through the hole, and the panels shall be separated by a glass spacer. The panels shall be free of corrosion products at the start of the test, and shall not be handled with the bare hands throughout the test.
2. The panels shall be degreased with acetone.
3. The panels shall be air dried at room temperature for not less than 15 minutes.
4. The panels shall be weighed to the nearest milligram.
5. The panels shall be dipped for not less than one minute in 1500 ml of aqueous solution consisting of 0.05N hydrobromic acid and 0.10N sulfuric acid at 200 °F. The aqueous solution shall be contained in a 3000 ml Pyrex or Vycor Griffin beaker and shall completely cover the test panels.
6. The panels shall be raised and suspended one inch above the hot solution for 15 minutes. The beaker shall be covered to minimize loss of vapor.
7. Steps 5 and 6 shall be repeated and the panels placed in a muffle furnace at 1100 °F for one hour.
8. The panels shall be removed from the furnace and allowed to cool to room temperature.
9. The panels shall be brushed with a brass brush to remove loose corrosion products.
10. Steps 5 through 9 shall be repeated three more times.
11. Steps 2, 3, and 4 shall be repeated.
12. A difference in weight loss between the two test panels greater than 3 grams per square foot indicates that the test is inconclusive and should be repeated.
13. Weight loss shall be multiplied by 12 to convert weight loss to grams per square foot.
14. A weight loss exceeding 8 grams per square foot shall constitute failure of this test.

Notes:

a. The aqueous solution shall be changed every four cycles.
b. Duplicates of active metals (zinc, iron) shall not be run in the same corrosive bath.
c. Iron and zinc may be run together if desired.
d. A minimum specific ratio of 25 ml of the corrosive medium per square inch of specimen surface should be used.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

113.	Contents not immobilized as specified for level A (see 5.1.1).
114.	Box not closed and sealed as specified for level A (see 5.1.1).
115.	Preservation not as specified for level C (see 5.1.2).
116.	Quantities packed together exceed the size or weight limitation of the box for level A or B (see 5.2.1 and 5.2.2).
117.	Strapping not as specified for level A (see 5.2.1).
118.	Packing not in accordance with the referenced document as specified for level C (see 5.2.3).
119.	Marking missing, illegible, incorrect, or incomplete (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or level C as specified (see 6.2).

5.1.1 Level A. Each muffler shall be placed in a close-fitting, end-opening box, conforming to PPP-B-636, class weather-resistant, grade as applicable to the weight of contents, style FOL. The contents shall be immobilized within the box in a manner to prevent movement or damage. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Level C. Each muffler shall be preserved in accordance with MIL-STD-1190.

5.2 Packing. Packing shall be level A, level B or level C as specified (see 6.2).

5.2.1 Level A. Mufflers of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. Mufflers of like description, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, style optional or boxes conforming to PPP-B-640, class 2, style optional, may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure shall be in accordance with the appendix to the applicable box specification.

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5.2.3 Level C. Mufflers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking specified by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mufflers are intended to reduce the exhaust noise level of internal combustion engines. The recommended maximum sound levels, shown in MIL-STD-1474, table VI, may be used as a guide in the selection of mufflers for use in various areas.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2, 4.3 and 6.3).
d. When the end caps and shell shall be welded (see 3.7).
e. When the muffler shall be painted (see 3.10).
f. Levels of preservation and packing required (see 5.1 and 5.2).
g. Any special marking required (see 5.3).

6.3 First article. When a first article inspection is required, the muffler should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Exhaust pipe
Noise reducer
Resonator
Silencer

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Army - CE

Preparing activity:
Army - ME

Project 2990-A004

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-525908(ME)	2. DOCUMENT DATE (YYMMDD) 911017

3. DOCUMENT TITLE
Mufflers, Exhaust, Internal Combustion Engine, General Purpose

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806	(2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center TTN: STRBE-TSE ort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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